

## [D.3.2] Universal AI: AIXI

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# The big picture: one agent, two halves

- **Universal AI** asks: what would an *optimally intelligent* agent look like, given unlimited compute and the weakest possible assumptions?
- It splits into two problems:
  - ① **Prediction (Solomonoff)**: passively observe a sequence  $x_1, x_2, \dots$  and predict the next symbol. *How should you learn?*
  - ② **Action (AIXI)**: interact with an unknown environment, take actions, collect reward. *How should you act?*
- **The common engine**: a single **Bayesian mixture**  $\xi$  over a class  $\mathcal{M}$  of candidate environments, weighted by a prior  $w_\nu$ .
  - Solomonoff:  $\xi$  (eventually) predicts as well as the truth  $\mu$ .
  - AIXI: the Bayes-optimal policy  $\pi_\xi^*$  (eventually) acts as well as the optimal policy  $\pi_\mu^*$  for true environment  $\mu$ .
- **Universal** choice:  $\mathcal{M}$  = all computable environments, prior  $w_\nu = 2^{-K(\nu)}$  ( $K$  is the “complexity” of the environment). Assumption “ $\mu \in \mathcal{M}$ ” becomes “*the universe is computable*”.

# The Bayesian mixture $\xi$

## Definition

Class  $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$  with prior weights  $w_\nu > 0$ ,  $\sum_\nu w_\nu = 1$ .

$$\xi(x_{1:t}) := \sum_{\nu \in \mathcal{M}} w_\nu \nu(x_{1:t}).$$

- **Key fact (mixture is a predictor):** its one-step prediction is a *posterior-weighted* average of the members' predictions (you prove this today!)

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid x_{<t}) \nu(x_t \mid x_{<t}), \quad w(\nu \mid x_{<t}) = w_\nu \frac{\nu(x_{<t})}{\xi(x_{<t})}.$$

- **Posterior update is multiplicative:**  $w(\nu \mid x_{1:t}) = w(\nu \mid x_{<t}) \cdot \frac{\nu(x_t \mid x_{<t})}{\xi(x_t \mid x_{<t})}$ : we reweight by how well  $\nu$  predicted vs.  $\xi$  (Bayes' rule!)

# From prediction to action

- Now the agent doesn't just watch: it **acts**, and its actions shape what it sees.
- At each step  $t$ : the agent picks action  $a_t$ , the environment returns a **percept**  $e_t = (o_t, r_t) = \text{observation} + \text{reward}$ .

$$h = a_1 e_1 a_2 e_2 a_3 e_3 \cdots \quad (\text{history } \mathfrak{a}_{<t})$$

- Spaces: actions  $\mathcal{A}$ , observations  $\mathcal{O}$ , rewards  $\mathcal{R} \subset [0, 1]$ , percepts  $\mathcal{E} = \mathcal{O} \times \mathcal{R}$ , and (finite) histories  $\mathcal{H}^* := (\mathcal{A} \times \mathcal{E})^* := \cup_{t=0}^{\infty} (\mathcal{A} \times \mathcal{E})^t$ .
- **No Markov / MDP assumption!** Both the policy and environment can depend on all prior percepts.

# The three players

## Policy $\pi$

$\pi(a_t \mid \mathfrak{a}_{<t}) : \mathcal{H} \rightarrow \Delta\mathcal{A}$  is the agent's distribution over the next action given history.

## Environment $\nu$

$\nu(e_t \mid \mathfrak{a}_{<t} a_t) : \mathcal{H} \times \mathcal{A} \rightarrow \Delta\mathcal{E}$  is the environment's distribution over the next percept given history and action. True (unknown) environment is  $\mu$ .

## Interaction measure $\nu^\pi$

Run  $\pi$  inside  $\nu$  and multiply along the history:

$$\nu^\pi(\mathfrak{a}_{t:m} \mid \mathfrak{a}_{<t}) := \prod_{k=t}^m \pi(a_k \mid \mathfrak{a}_{<k}) \nu(e_k \mid \mathfrak{a}_{<k} a_k).$$

## Value function (assuming geometric discount $\gamma$ )

$$\begin{aligned} V_{\nu}^{\pi,m}(\mathfrak{a}_{<t}) &= (1 - \gamma) \mathbb{E}_{\nu}^{\pi}[G_{t-1:m} \mid \mathfrak{a}_{<t}] \text{ with } G_{t-1:m} = \sum_{k=t}^m \gamma^{k-t} r_k \\ &= (1 - \gamma) \sum_{\mathfrak{a}_{t:m}} \nu^{\pi}(\mathfrak{a}_{t:m} \mid \mathfrak{a}_{<t}) \sum_{k=t}^m \gamma^{k-t} r_k \end{aligned}$$

The **infinite-horizon value** is defined as the pointwise limit

$$V_{\nu}^{\pi} \equiv V_{\nu}^{\pi,\infty}(\mathfrak{a}_{<t}) := \lim_{m \rightarrow \infty} V_{\nu}^{\pi,m}(\mathfrak{a}_{<t}).$$

The **optimal value** is  $V_{\nu}^{*,m}(\mathfrak{a}_{<t}) := \sup_{\pi} V_{\nu}^{\pi,m}(\mathfrak{a}_{<t})$ , with  $V_{\nu}^* \equiv V_{\nu}^{*,\infty}$ .

- **Optimal value**  $V_{\nu}^* := \sup_{\pi} V_{\nu}^{\pi}$ ; an optimal policy  $\pi_{\nu}^*$  satisfies  $V_{\nu}^{\pi_{\nu}^*} = V_{\nu}^*$ .
- **Bayes-optimal policy**  $\pi_{\xi}^* \in \arg \max_{\pi} V_{\xi}^{\pi}$ : acts optimally w.r.t. the *mixture*  $\xi$ .

## AIXI

$\pi_{\xi}^*$  with  $\mathcal{M}$  = all  $\approx$ computable environments<sup>a</sup> and prior  $w_{\nu} = 2^{-K(\nu)}$ . Write  $\xi_U$  for  $\xi$  with this choice of  $\mathcal{M}$  and  $w_{\nu}$ , so  $\text{AIXI} = \pi_{\xi_U}^*$ .

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<sup>a</sup>lower-semicomputable semi-measures

**Intuition:** *Act optimally according to environments that you believe to be in, assuming the simplest environments consistent with observations are more likely to be true.*

## AIXI, explicitly

max over actions,  $\sum$  over percepts, choosing actions that maximise the  $\xi$ -weighted discounted return:

$$a_t = \underset{a_t}{\text{arg max}} \lim_{m \rightarrow \infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} \mid \mathfrak{a}_{<t} a_{t:m}) \sum_{k=t}^m \gamma^{k-t} r_k$$

# Result 1: On-policy value convergence

## Theorem

For any  $\mu \in \mathcal{M}$  and any policy  $\pi$ ,

$$V_{\xi}^{\pi}(\mathfrak{a}_{<t}) - V_{\mu}^{\pi}(\mathfrak{a}_{<t}) \rightarrow 0 \quad \mu^{\pi}\text{-almost surely.}$$

**Intuition:** *Following policy  $\pi$ , pretending the environment is  $\xi$ , eventually will act as if we knew the true environment  $\mu$ .*

## Result 2: AIXI can't be fooled

### Theorem (deterministic $\mu$ )

If  $V_{\mu}^*(\mathfrak{a}_{<t}) > \varepsilon$  along history  $\mathfrak{a}_{<t}$ , then

$$V_{\xi}^*(\mathfrak{a}_{<t}) \geq w_{\mu} \varepsilon > 0.$$

**Intuition:** *Whenever good behaviour in  $\mu$  is achievable, the Bayes-optimal agent secures at least a fraction proportional to the prior weight of  $\mu$ .*

## Result 3: Self-optimizing property (stretch goal)

### Theorem

If *some* policy can learn to act optimally in every  $\nu \in \mathcal{M}$ , then  $\pi_\xi^*$  does too:

$$V_\mu^*(\mathfrak{x}_{<t}) - V_\mu^{\pi_\xi^*}(\mathfrak{x}_{<t}) \rightarrow 0 \quad \mu^\pi\text{-a.s.}$$

**Intuition:** *AIXI learns to play as well as the optimal agent that knows the true environment  $\mu$ .*

# Takeaways

- **Solomonoff** solves induction: mix over all computable hypotheses, weighted by simplicity. Total prediction error  $\leq K(\mu) \ln 2$ .
- **AIXI** lifts this to action: the Bayes-optimal policy over the same universal mixture. It learns to predict *and* to act as well as if it knew the world.
- **The catch:** both are *incomputable* ( $K$  is incomputable;  $\mathcal{M}$  is intractable to enumerate, the expectimax is intractable). They are gold standards, not computable algorithms, and the prior's  $2^{-K(\nu)}$  hides a dependency on the choice of universal Turing machine.
- **Why care for safety:** a precise, assumption-light model of idealized intelligence providing a lens on what optimal agents do. For example, we can show under what circumstances AIXI would [wirehead](#) [RO11],
- **Approximations:** [MC-AIXI-CTW](#) [VNH<sup>+</sup>11] gives an approximation using Context Tree Weighting to replace  $\xi$  and Monte-Carlo-Tree-Search to replace the expectimax. [safely rather than recklessly explore](#) [CCH20], or [self-modify](#) [OR11] as a model of what superintelligent agents would do.

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